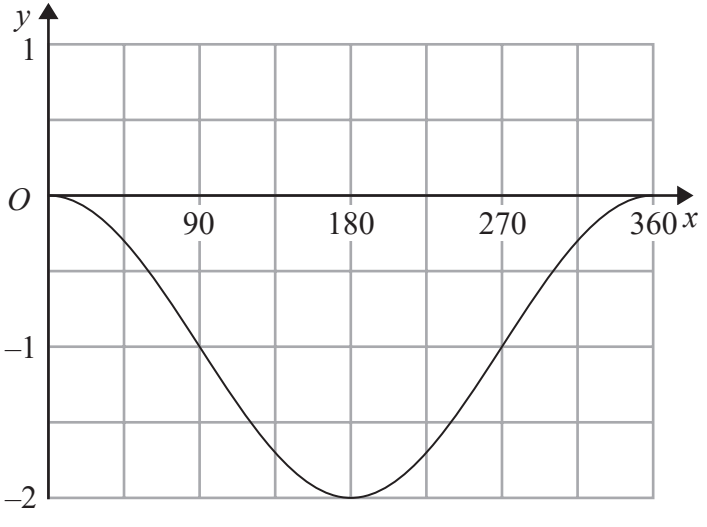
18 Here is a sketch of the curve $y = \sin(x + a)° + b$



1 mark
a=90
Maximum at x=0

1 mark
b=−1
Midline of0 and−2

Given that $0 < a < 360$
find the value of a and the value of b.

Sine shifted 90° left: a=90.

$a =$ 90

Midline:(0 + (−2)) ÷ 2 = −1.

$b =$ −1

(Total for Question 18 is 2 marks)